

PRAMITHAS UPRETI

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EDUCATION

Colby College, Waterville, ME

Computer Science & Mathematical Sciences Double Major with a minor in Philosophy

4-time Dean's List

Sample Coursework: Algorithm Design & Analysis, Algorithmic Game Theory, Software Engineering, Data Structures & Algorithms, Data Visualization, AI: Computer Vision, Programming Languages, Real Analysis, Linear Algebra, Mathematical Reasoning, Mathematical Modeling, Differential Equations

University of Oxford, Oxford, UK

Visiting Non-Matriculating Student of Mathematics & Philosophy

Sample Coursework: Probability & Ethics

Bachelor of Arts 2026

Cumulative GPA: 3.90

Oct 2024-Mar 2025

EXPERIENCE

Data Analytics & Research Intern, International Sibling Society, Las Vegas, Nevada

Jun 2026-Aug 2026

- Spearheaded the design & implementation of scalable data collection frameworks in Python to monitor user acquisition, engagement, retention, & platform performance, enabling evidence-based evaluation of organizational impact.
- Developed interactive dashboards, KPI reporting systems, & advanced data visualizations that transform complex datasets into actionable insights, supporting strategic decision-making, performance optimization, & A/B testing initiatives.
- Partnered with cross-functional teams across product, research, & operations to leverage statistical analysis, impact measurement methodologies, & data-driven experimentation to assess & enhance the effectiveness of a global nonprofit platform serving stakeholders in 190+ countries.

Research Assistant, Colby College, Waterville, Maine

May 2024-Aug 2024

- Conducted research on dynamic pricing algorithms in the context of the online set cover problem, focusing on the client-server model with resource-specific pricing constraints.
- Developed online algorithms adaptable to dynamic pricing, analyzing their efficiency & applicability.
- Collaborated to structure theoretical frameworks & evaluate algorithmic efficiency in real-time client selection scenarios then authored documentation on algorithmic approaches & performance analysis.

Research Assistant, Colby College, Waterville, Maine

Jun 2023-Aug 2023

- Conducted comprehensive bibliometric analysis of 5 decades of ice-core science, mapping research trends & influences.
- Developed a Guided Machine Learning model using deep learning frameworks like PyTorch & TensorFlow to make predictive analysis in climate research applications.
- Worked on dynamic topic classification & geographic distribution within ice-core science research.

Teaching Assistant, Colby CTL, Waterville, Maine

Dec 2022-Present

- Provided targeted instructions to students across seven courses (CS333, CS252, CS231, CS152, MA130, MA120, & PL151) in computer science, mathematics, & philosophy.
- Collaborated with professors to grade assignments & quizzes; led help sessions to clarify course concepts & improve students' logical understanding.

PUBLICATION

Dynamic Pricing Algorithms for Online Set Cover. Bender, M., Desai, A., He, J., Thompson, O., Upreti, P. (2024). arXiv preprint arXiv:2409.15094. Available at <https://arxiv.org/abs/2409.15094>

PROJECTS

- Logic Visualizer | JavaScript, HTML, CSS** | Developed an interactive web-based logic visualization platform for constructing & evaluating propositional formulas & logical circuits in real time. Designed dynamic visual representations of logical operators & truth-functional relationships, enabling intuitive exploration of formal reasoning processes. Implemented client-side parsing, state management, & live circuit evaluation to provide immediate feedback as users modified logical expressions. Engineered an educational interface that bridges symbolic logic & computational visualization, making abstract logical structures accessible through interactive graphics.
- Semantic Logic Editor | React, TypeScript, Vite, Knowledge Representation, Formal Semantics** | Developed an interactive platform for semantic authoring & formal reasoning, transforming mathematical prose into structured definitions, theorems, proofs, & dependency graphs. Implemented document parsing, logical diagnostics, & formal-export pipelines supporting knowledge representation & proof-assistant workflows. Engineered a React/TypeScript application using Vite & GitHub Pages with dynamic visualization of logical relationships & AI-oriented formalization infrastructure. Built interactive visualizations of logical dependencies & conceptual relationships, allowing users to explore the structure of mathematical arguments & formal theories through dynamic graph-based interfaces.
- Quantum Particle Simulator | Python** | Designed & built a robust simulation framework that models the probabilistic behavior & wave function of a single quantum particle in a bounded one-dimensional potential. Achieved realistic quantum state representations & validated results against standard quantum mechanical principles.
- AI Trash Detection Model | Python, Jupyter** | Developed a series of AI models using Computer Vision, TensorFlow, Keras & Hugging Face transformers for trash identification across complex terrains as a part of a group. Collaborated with a research team to analyze & map trash accumulation on Allen Island, contributing to environmental studies.

ACTIVITIES

Carnegie Mellon University, Summer School, Pittsburgh, Pennsylvania

Jun 2026

- Attended the summer program in Logic & Formal Epistemology. Topics include Epistemic Logic & Topology, Decision Theory & AI Agent Foundations, & Gavagai to a full grammar.

Roux Institute at Northeastern University, AI Camp, Portland, Maine

Jul 2023

- Collaborated to apply ML models for data analysis, utilizing hands-on expertise with data-driven AI solutions.